



Creating Realistic Materials for Blender Projects

Having completed the three-part series, 'Animate Using Blender', this article focuses on the science behind the process of creating good materials and textures within Blender.

If modelling and rigging are the most significant aspects of the final animation film that we all enjoy to watch, no less important is the process of providing good materials and textures to objects created within Blender. Any object or animation clip created in Blender (or in any good animation package) loses half its charm when it is not given appropriate materials and textures. Instead of being a life-like rendition, it takes on a plastic toy-like look. Realism is the key to success in the animation industry. How realistically you create life-like motion is the key to getting to the top. On the other hand, if not coupled with realism in appearance, the entire effort of aiming to be the best is wasted. Hence, it is important to create good materials for your projects.

Light and objects in the physical world

We need to understand light and its interaction with objects in the physical world. Though its composition is not totally understood, it plays a great role in our world. It makes objects visible to us; when it strikes a substance, light either refracts, reflects, or is absorbed, based on the properties of the material the object is made of. Reflection is the bouncing back of light after it hits an opaque object. The extent of regular reflection

depends on the optical smoothness of a medium. Refraction is the bending of light when it traverses from one medium to another, at the plane of separation. The extent of bending depends upon the optical density of the medium into which the light travels. The phenomena of reflection and refraction are important to understand real-world surfaces.

Blender simulates real-life materials

Blender simulates how real-life materials interact with lights. In the physical world, the extent of a material's shine is determined by the microscopic grooves and bumps on its surface, which scatter the incoming light over wider or smaller angles. The rougher the surface is, the wider the angle the incident light will be reflected at, causing blurrier highlights and reflections. In Blender's ray-tracer, this process has been incorporated—both for perfectly glossy mirror-like (or clear glass-like) surfaces, and also for less glossy surfaces causing blurry reflections or refractions. Rather than taking a single reflection sample for fully glossy surfaces, blurry ray-tracing works by blending multiple samples, distributed in a cone shape around the outward ray direction. The more